

Eigenvalues and inverse problems

Finitely many unitary classes with prescribed shifted singular values

Difficulty: challenging **Importance:** interesting to the community

Status: Partially resolved

Topic: Inverse determination from complete pseudospectral data

Rating rationale: Generic finiteness follows from invariant theory, but the exceptional fibers resist that argument; the question measures how much complete pseudospectral data can identify a nonnormal matrix.

Context and notation

For a complex $N \times N$ matrix M , write $s_1(M) \geq \dots \geq s_N(M)$ for its singular values. Say $A \sim_{\text{sip}} B$ if

$$s_j(A - zI) = s_j(B - zI) \quad (z \in \mathbb{C}, 1 \leq j \leq N).$$

This is equality of *super-identical pseudospectral* data: it includes every singular value of every scalar shift.

Problem statement

Is it true that for every $N \geq 1$ there is an integer $M_N \geq 1$ such that every family $A_1, \dots, A_{M_N+1} \in \mathbb{C}^{N \times N}$ with $A_i \sim_{\text{sip}} A_j$ for all i, j contains a pair satisfying

$$A_j = U^* A_i U \quad \text{for some } i < j \text{ and } U^* U = I?$$

Equivalently, is every such data fiber a union of at most M_N unitary similarity classes, with a bound depending only on N ?

References

- Maxime Fortier Bourque and Thomas Ransford, *Super-identical pseudospectra*, *Journal of the London Mathematical Society* **79**(2), 511–528 (2009), **Theorem 1.4 and §6.2**. The theorem removes a closed measure-zero exceptional set; the question asks to remove this exception. [Author manuscript](#).
- Thomas Ransford, *Pseudospectra and matrix behaviour*, **Theorem 5.4 and the following discussion, pp. 10–11**. This restates the finiteness theorem and explicitly leaves the empty-exceptional-set case open. It explains why replacing algebraicity with integrality fails to prove it.
- G. Armentia, J. M. Gracia and F. E. Velasco, *Identical pseudospectra of any geometric multiplicity*, *Linear Algebra and its Applications* **436**(6), 1683–1688 (2012). Their similarity theorem for super-identical pseudospectra gives ordinary similarity, which allows nonunitary changes of basis and does not answer the displayed question.

The original low-dimensional results give $M_2 = 1$ and $M_3 = 2$; $M_1 = 1$ is immediate. The unresolved question concerns exceptional data fibers in higher dimensions, rather than generic matrices.

Scope and status check

The quantifiers above are an explicit restatement of the source question about the exceptional set, with M_N counting classes instead of the source's pigeonhole family size. The source is older: current openness rests on bounded later-literature searches, not a recent explicit reaffirmation. Searches through 2026-09-11 for super-identical pseudospectra, finite unitary classes, the exceptional set and related similarity results found no resolution. The 2012 ordinary-similarity theorem and the 2022 norm estimates below were checked for scope. They do not remove the exceptional set.